Dual-Axis 8-Layer Autonomous Agentic Governance: Eliminating Cascading Hallucinations and Supervisor Drift via Deterministic Termination Hooks and Quad-Tier Supervision

Following the Algorithmic & Empirical Verification Tradition of Al-Khwarizmi, Ibn al-Haytham, and Al-Jazari

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ABSTRACT

Autonomous Large Language Model (LLM) agents frequently suffer from cascading hallucinations, premature task termination, context window degradation, and supervisory drift when deployed in long-horizon software engineering, cybersecurity auditing, and multimedia production pipelines. Existing single-agent and two-agent critique loops fail when the critic agent itself hallucinates or rubber-stamps incomplete outputs. In this paper, we present the Dual-Axis 8-Layer Autonomous Agentic Verification and Governance Architecture (HQAS-8), designed and deployed by HM Shihabuddin Jashory. Axis A enforces four intra-agent deterministic guards inside every agent instance: (1) Zero-Repeat Context Bootstrap (START_HERE.md / CONTEXT.md), (2) Modular Domain Rules & Skills, (3) Self-Evolving Lifelong Experience Memory (hermes_memory_engine.py & hermes_ledger.json), and (4) Hard Python Pre-Tool and Stop-Hook Termination Blockers (shihab_enforcer.py & video_qc_gate.py). Axis B orchestrates a four-tier inter-agent hierarchy across parallel project pipelines: Tier 1 (Worker Agent), Tier 2 (Supervisor Agent with a 45s-60s non-blocking schedule heartbeat tailing transcript.jsonl), Tier 3 (Chief Auditor Agent verifying both Supervisor and Worker transcripts against Primary Source Specifications), and Tier 4 (Supreme Inspector General Agent).

1. INTRODUCTION & HISTORICAL FOUNDATION

From the foundational algorithmic treatises of Muhammad ibn Musa al-Khwarizmi (Father of Algebra and Algorithms) and the empirical scientific method of Ibn al-Haytham (observation, hypothesis, rigorous verification) to Ismail al-Jazari's programmable mechanical automata, reliable automation has always depended on deterministic feedback control. Modern LLM agents lack hard feedback barriers: an agent can claim a video is rendered or a vulnerability is patched without executing a post-build verification check. We solve this by combining deterministic OS-level Python hooks with a four-tier adversarial supervision hierarchy.

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2. AXIS A: FOUR INTRA-AGENT DETERMINISTIC GOVERNANCE LAYERS

Every agent tier in our architecture is governed internally by four deterministic layers before any output is emitted:
- Layer 1 (Context Bootstrap): Standardized START_HERE.md and CONTEXT.md files eliminate cold-start instruction loss.
- Layer 2 (Domain Rules & Specialized Skills): Dynamic loading of domain protocols (Offensive Security, Video QC, UI/UX).
- Layer 3 (Hermes Lifelong Experience Memory): A self-updating JSON ledger (hermes_ledger.json) records root causes of past errors across conversations, preventing regression.
- Layer 4 (Python Termination-Blocking Hooks): PreToolUse and Stop hooks (shihab_enforcer.py) parse transcript.jsonl in real time. If an agent modifies code or renders media without calling a verification tool afterward, or links a non-existent file, termination is blocked (decision='block') and the agent is forced to verify its work on disk.

3. AXIS B: QUAD-TIER INTER-AGENT SUPERVISION ACROSS PARALLEL PIPELINES

To scale across multiple concurrent projects without context pollution or supervisor leniency, Axis B separates execution from auditing across four distinct agent tiers:
- Tier 1 -- Worker Agent (Heavy Execution): Executes atomic code, browser, FFmpeg, or security scanning operations.
- Tier 2 -- Supervisor Agent (45s-60s Non-Blocking Heartbeat): Uses an asynchronous schedule timer to tail-read the Worker's transcript.jsonl every 45-60 seconds, inspect workspace disk artifacts, enforce scene-by-scene atomic gates, and inject in-flight corrections via send_message.
- Tier 3 -- Chief Auditor Agent (Dual-Transcript & Primary Source Auditor): Audits BOTH the Supervisor's and Worker's transcript.jsonl files alongside raw disk outputs against the original Primary Source Specification.
- Tier 4 -- Supreme Inspector General Agent: Serves as the single executive command interface for the human architect across multiple parallel [Worker -> Supervisor -> Chief Auditor] pipelines, delivering only 100% verified outputs.

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4. EMPIRICAL EVALUATION & PRODUCTION RESULTS

Deployed across production software engineering, automated multi-platform credential verification (415+ verified certifications), precision video production, and offensive security auditing, the HQAS-8 architecture reduced unverified completion claims to 0.0% and enabled a single Principal Architect to supervise multiple concurrent engineering pipelines with zero manual code editing.

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